Member 4: AI-IoT Engineer Task – Smart Agriculture Proposal

Title: SmartAgri – AI-Enabled IoT System for Crop Yield Optimization

Problem Statement:

Farmers in many regions struggle with unpredictable crop yields due to climate variability and inefficient resource use. A smart system is needed to provide data-driven insights to optimize farming decisions in real time.

System Description:

The SmartAgri system integrates IoT sensors with AI to monitor environmental conditions and predict crop yields. By using real-time sensor inputs, the system forecasts outcomes and guides decisions such as irrigation and fertilization.

Sensors Required:

• Soil Moisture Sensor – monitors soil water levels

• Temperature Sensor – detects ambient climate

• Humidity Sensor – tracks environmental humidity

• pH Sensor – monitors soil acidity

• Light Sensor – measures solar radiation

• (Optional) NDVI Camera or Drone – for crop health monitoring

AI Model:

• Model Type: Supervised learning (e.g., Random Forest or LSTM regression model)
• Inputs: Time-series data from IoT sensors + historical yield records
• Output: Predicted crop yield in kg/ha or tons/acre
• Frameworks: TensorFlow, scikit-learn, PyTorch

Key Features:

• Real-time alerts for irrigation and fertilization

• Risk detection (e.g., drought, pest outbreaks)

• Visual dashboard for data and yield forecast display

Benefits:

• Improved crop yield predictions

- Efficient resource allocation
- Early detection of anomalies or risks

Data Flow Diagram